



SWAYAM NPTEL COURSE ON STATISTICAL FOUNDATION FOR BIG DATA ANALYSIS

BY

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Module 10: Sample Principal Component Analysis

Lecture 01: Population vs Sample Principal Component Analysis



Sample Setup and Notation

Sample Observations

Let

$$X^{(1)}, X^{(2)}, \dots, X^{(n)} \in \mathbb{R}^p$$

be an i.i.d. sample a given population.

Sample Mean and Covariance

Sample mean:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X^{(i)}.$$

Sample covariance matrix:

$$\hat{\Sigma} = \frac{1}{n-1} \sum_{i=1}^n (X^{(i)} - \bar{X})(X^{(i)} - \bar{X})^T.$$

Interpretation

Sample PCA analyzes variability present in a finite dataset as an empirical estimate of population PCA.

Goal of Sample PCA

Goal

Sample PCA seeks orthogonal linear combinations of the variables that capture the largest possible sample variance, using the sample covariance matrix $\hat{\Sigma}$.

Sample PCA: Setup

Centered Data

Define centered observations:

$$Z^{(i)} = X^{(i)} - \bar{X}.$$

Definition of Components

Sample principal components are:

$$\hat{Y}_k^{(i)} = q_k^T Z^{(i)}, \quad k = 1, \dots, p.$$

Goal

Find orthogonal directions q_k that:

- maximize the sample variance of $\hat{Y}_k$
- ensure the components are sample uncorrelated

Properties of Sample PCA

Variance

$$\text{Var}_n(\hat{Y}_k) = q_k^T \hat{\Sigma} q_k.$$

Uncorrelatedness

$$\text{Cov}_n(\hat{Y}_i, \hat{Y}_j) = q_i^T \hat{\Sigma} q_j = 0, \quad i \neq j.$$

Interpretation

The vectors q_k rotate the coordinate system so that the components $\hat{Y}_k$ are uncorrelated and ordered by decreasing sample variance.

First Principal Component

Optimization Problem

$$q_1 = \arg \max_{q \in \mathbb{R}^p} q^T \hat{\Sigma} q \quad \text{subject to } q^T q = 1.$$

Component

$$\hat{Y}_1^{(i)} = q_1^T Z^{(i)}.$$

Meaning

This direction captures the maximum possible sample variance among all unit-length linear combinations of $Z^{(i)}$.

Second Principal Component

Optimization Problem

$$q_2 = \arg \max_{q \in \mathbb{R}^p} q^T \hat{\Sigma} q \quad \text{subject to } q^T q = 1, \quad q^T q_1 = 0.$$

Component

$$\hat{Y}_2^{(i)} = q_2^T Z^{(i)}.$$

Uncorrelatedness Constraint

The constraint $q^T q_1 = 0$ ensures

$$\text{Cov}_n(\hat{Y}_1, \hat{Y}_2) = 0.$$

General k -th Principal Component

Optimization Problem

$$q_k = \arg \max_{q \in \mathbb{R}^p} q^T \hat{\Sigma} q$$

subject to

$$q^T q = 1, \quad q^T q_j = 0 \text{ for all } j < k.$$

Component

$$\hat{Y}_k^{(i)} = q_k^T Z^{(i)}.$$

Interpretation

Each new component maximizes sample variance under the requirement of being uncorrelated with all previous components.

Interpretation of Sample PCA

Geometric View

Sample PCA constructs a new coordinate system aligned with directions of maximal variability in the sample.

Transformation

The vectors $q_1, \dots, q_p$ form an orthogonal rotation:

$$\hat{Y}_k = q_k^T (X - \bar{X}).$$

Key Insight

Sample PCA is an empirical approximation of population PCA, based on the finite-sample covariance structure.

Why Do We Need Sample PCA?

Population Quantities Are Unknown

Population PCA requires the true mean μ and covariance matrix Σ , but these are never observed in practice. We only have access to a finite sample of observations.

Population Relationships Are Not Directly Observable

Correlations and dependence structures in the population cannot be known without data. Sample PCA uses the observed sample to infer these hidden relationships.

Data-Driven Discovery

Sample PCA reveals:

- correlations between variables,
- directions of maximal variability,
- low-dimensional structure,
- patterns not visible at the population level.

Role and Advantages of Sample PCA

Approximation of Population PCA

As sample size grows:

$$\bar{X} \rightarrow \mu, \quad \hat{\Sigma} \rightarrow \Sigma,$$

so sample PCA becomes a consistent estimate of population PCA.

Necessary in Real Applications

In real data problems, the full population is unobservable. Thus sample PCA is the only feasible way to perform PCA.

Handling Complex Structure

Real populations may have:

- unknown or complicated relationships,
- noise or nonlinear dependencies,
- high-dimensional interactions.

Sample PCA provides a model-free method to extract structure from data.

Population PCA vs. Sample PCA: Overview

Population PCA

Population PCA is defined using the **true** distribution of the random vector X . It relies on:

$$\mu = \mathbb{E}[X], \quad \Sigma = \text{Cov}(X).$$

The principal components are the eigenvectors of the population covariance matrix Σ .

Sample PCA

Sample PCA is computed from the **observed data**:

$$X^{(1)}, \dots, X^{(n)}.$$

It uses:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X^{(i)}, \quad \hat{\Sigma} = \frac{1}{n-1} \sum_{i=1}^n (X^{(i)} - \bar{X})(X^{(i)} - \bar{X})^T.$$

The sample principal components are eigenvectors of $\hat{\Sigma}$.

Why Population and Sample PCA Differ

Unknown Population Structure

In practice, μ and Σ are **unknown**. We only observe a finite dataset, so population PCA is usually **inaccessible**. Sample PCA provides an estimate based on observed data.

Sampling Variability

Because the dataset is finite:

- $\bar{X} \neq \mu$,
- $\hat{\Sigma} \neq \Sigma$,
- sample eigenvalues $\hat{\lambda}_k$ fluctuate,
- normalized eigenvectors of sample q_k differ from normalized eigenvectors of the population.

Thus the directions of maximal variation differ between population and sample PCA.

Interpretation Difference

Population PCA describes the **true** variability of the distribution. Sample PCA describes the variability **present in the data**, which includes noise.

Connections Between Population and Sample PCA

Convergence as $n \rightarrow \infty$

As the sample size grows:

$$\bar{X} \rightarrow \mu, \quad \hat{\Sigma} \rightarrow \Sigma,$$

and therefore:

$$\hat{\lambda}_k \rightarrow \lambda_k, \quad q_k^{(\text{sample})} \rightarrow q_k^{(\text{population})}.$$

Sample PCA becomes a consistent estimator of population PCA.

Practical Importance

- Population PCA is mostly theoretical—requires knowing Σ .
- Sample PCA is used in every real application.
- It uncovers structure that cannot be accessed at the population level.

Key Insight

Sample PCA is the **empirical approximation** of Population PCA, built from the data we observe.

Example 1

Goal

Using a simulation, we compare the **population PCA** of a known bivariate normal distribution with the **sample PCA** computed from random samples of sizes

$$n = 20, \quad 200, \quad 2000.$$

What We Show

- Population covariance ellipse
- Population principal components (PC1, PC2)
- Sample points
- Sample principal components

Population Model

Distribution

$$X \sim \mathcal{N}(\mu, \Sigma), \quad \mu = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 4 & 2 \\ 2 & 3 \end{pmatrix}.$$

Interpretation

- Σ determines the orientation and size of the covariance ellipse.
- PCA finds the eigenvectors and eigenvalues of Σ .

Population PCA

Eigen-Decomposition

$$\Sigma = Q\Lambda Q^T, \quad \Lambda = \text{diag}(\lambda_1, \lambda_2), \quad Q = [q_1 \ q_2].$$

Meaning

- q_1 : direction of maximum variance (Population PC1)
- q_2 : orthogonal direction with smaller variance (Population PC2)
- Eigenvalues λ_1, λ_2 determine ellipse axis lengths.

Sampling Procedure

Generate Random Sample

$$X_1, \dots, X_n \stackrel{iid}{\sim} \mathcal{N}(\mu, \Sigma)$$

Sample Statistics

Sample mean:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

Centered observations:

$$Y_i = X_i - \bar{X}$$

Sample covariance:

$$\hat{\Sigma} = \frac{1}{n-1} \sum_{i=1}^n Y_i Y_i^\top.$$

Sample PCA

Eigen-Decomposition

$$\hat{\Sigma} = \hat{Q} \hat{\Lambda} \hat{Q}^T,$$

where $\hat{Q} = [\hat{q}_1 \ \hat{q}_2]$.

Interpretation

- $\hat{q}_1$: estimated PC1 direction
- $\hat{q}_2$: estimated PC2 direction
- Sample PCs differ from population PCs due to sampling variability.

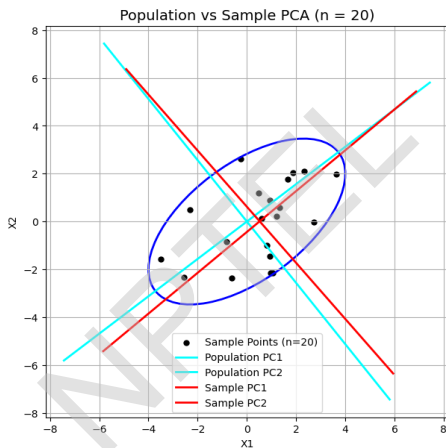
Each Figure Shows

- Random sample of points
- Population covariance ellipse
- Population PC1 and PC2 (cyan lines)
- Sample PC1 and PC2 (red lines)

Purpose

To compare how well the sample PCA matches the true population PCA for different sample sizes.

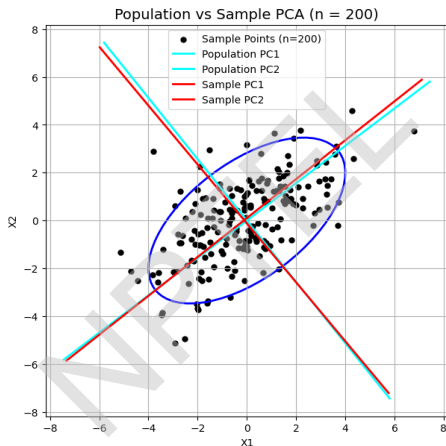
Case $n = 20$



Observations

- Substantial sampling variability
- Sample mean differs noticeably from μ
- Sample PCs visibly rotated relative to population PCs

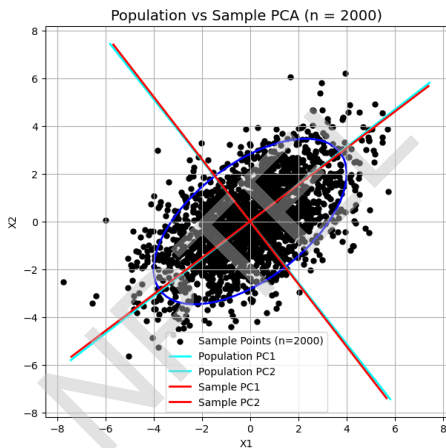
Case $n = 200$



Observations

- Sample covariance resembles the population covariance
- PCs begin to align with the population directions

Case $n = 2000$



Observations

- Very little sampling noise
- Sample PCA closely matches population PCA
- Sample and population PC lines nearly overlap

Summary

Key Points

- PCA reveals covariance structure: orientation and spread.
- Sample PCA estimates these from data.
- Small samples produce unstable PC estimates.
- Larger samples produce much more accurate PCA directions.

References

- ① Richard Arnold Johnson, and W. Dean Wichern. Applied multivariate statistical analysis, 2002.
- ② I. T. Jolliffe, Principal Component Analysis, Springer New York, January 2002.
- ③ David C Lay, Judi McDonald, and Steven R Lay Linear Algebra and Its Applications, 6th edition, Pearson (2020).

Thank you





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Module 10: Sample Principal Component Analysis

Lecture 02: Mathematical Formulation of Sample PCA



Sample PCA: Setup and Definitions

Sample Data

Let $X^{(1)}, \dots, X^{(n)} \in \mathbb{R}^p$ be i.i.d. observations from a given population. Define the sample mean:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X^{(i)}.$$

Sample Covariance Matrix

The unbiased sample covariance matrix is

$$S = \hat{\Sigma} = \frac{1}{n-1} \sum_{i=1}^n (X^{(i)} - \bar{X})(X^{(i)} - \bar{X})^T.$$

Sample Principal Components

Let $(\hat{\lambda}_i, \hat{q}_i)$ be the eigenvalue-normalized eigenvector pairs of S , with $\hat{\lambda}_1 \geq \dots \geq \hat{\lambda}_p \geq 0$. The sample principal components are

$$\hat{Y}_i = \hat{q}_i^T (X - \bar{X}), \quad i = 1, \dots, p.$$

Theorem 1: Sample PCs Are Uncorrelated and Have Variances Equal to Eigenvalues

Theorem

Let $(\hat{\lambda}_i, \hat{q}_i)$ be the eigenvalue-eigenvector pairs of the sample covariance matrix S . Then the sample principal components

$$\hat{Y}_i = \hat{q}_i^T (X - \bar{X}), \quad i = 1, \dots, p$$

satisfy:

- $\text{Var}(\hat{Y}_i) = \hat{\lambda}_i$
- $\text{Cov}(\hat{Y}_i, \hat{Y}_j) = 0$ for $i \neq j$

Proof of Theorem 1: Step 1 – Eigen Decomposition of Sample Covariance

Centered Data Matrix

Define the centered data matrix

$$X_c = \begin{pmatrix} (X^{(1)} - \bar{X})^T \\ (X^{(2)} - \bar{X})^T \\ \vdots \\ (X^{(n)} - \bar{X})^T \end{pmatrix}_{n \times p}.$$

Then the sample covariance matrix can be written as

$$S = \frac{1}{n-1} X_c^T X_c.$$

Eigenvalue Decomposition

Since S is symmetric, it can be diagonalized:

$$S = \hat{Q} \hat{\Lambda} \hat{Q}^T$$

where

- $\hat{Q} = [\hat{q}_1, \dots, \hat{q}_p]$ is an orthonormal matrix ($\hat{Q}^T \hat{Q} = I_p$),
- $\hat{\Lambda} = \text{diag}(\hat{\lambda}_1, \dots, \hat{\lambda}_p)$ contains eigenvalues of S .

Proof of Theorem 1: Step 2 – Variance of Sample PCs

Variance of $\hat{Y}_i$

$$\text{Var}(\hat{Y}_i) = \text{Var}(\hat{q}_i^T (X - \bar{X})) = \hat{q}_i^T S \hat{q}_i = \hat{q}_i^T (\hat{Q} \hat{\Lambda} \hat{Q}^T) \hat{q}_i = \hat{q}_i^T (\hat{\lambda}_i \hat{q}_i) = \hat{\lambda}_i.$$

Interpretation

The variance of each sample principal component is exactly the corresponding sample eigenvalue of S .

Proof of Theorem 1: Step 3 – Covariance Between Different Sample PCs

Covariance for $i \neq j$

$$\text{Cov}(\hat{Y}_i, \hat{Y}_j) = \text{Cov}(\hat{q}_i^T (X - \bar{X}), \hat{q}_j^T (X - \bar{X})) = \hat{q}_i^T S \hat{q}_j = \hat{q}_i^T (\hat{\lambda}_j \hat{q}_j) = \hat{\lambda}_j (\hat{q}_i^T \hat{q}_j) = 0$$

because $\hat{q}_i^T \hat{q}_j = 0$ for $i \neq j$ (orthonormality).

Interpretation

Sample PCs are uncorrelated by construction, mirroring the population PCA property.

Theorem 2: Proportion of Total Sample Variance Explained

Theorem

Let $\hat{\lambda}_1, \dots, \hat{\lambda}_p$ be eigenvalues of the sample covariance matrix S . Then

$$\text{Total Sample Variance} = \text{tr}(S) = \sum_{i=1}^p \hat{\lambda}_i$$

and the proportion of total sample variance explained by the k -th sample PC is

$$\frac{\hat{\lambda}_k}{\hat{\lambda}_1 + \dots + \hat{\lambda}_p}.$$

Proof of Theorem 2

Step 1: Total Sample Variance

The total variance of all variables in the sample is the sum of sample variances of the coordinates:

$$\text{Var}_{\text{total}}^{(\text{sample})} = \sum_{i=1}^p S_{ii} = \text{tr}(S).$$

Step 2: Eigenvalues Sum to Trace

For a symmetric matrix:

$$\text{tr}(S) = \sum_{i=1}^p \hat{\lambda}_i.$$

Thus the proportion of variance explained by the k -th component is

$$\frac{\hat{\lambda}_k}{\sum_{i=1}^p \hat{\lambda}_i}.$$

Theorem 3: Correlation Between Sample PCs and Original Variables

Theorem

For the sample PC $\hat{Y}_i = \hat{q}_i^T (X - \bar{X})$ and original variable X_k (with sample variance S_{kk}), the sample correlation is

$$\hat{\rho}_{\hat{Y}_i, X_k} = \frac{\hat{q}_{ik} \sqrt{\hat{\lambda}_i}}{\sqrt{S_{kk}}},$$

where $\hat{q}_{ik}$ is the k -th element of eigenvector $\hat{q}_i$.

Proof of Theorem 3

Step 1: Sample Covariance

$$\text{Cov}(\hat{Y}_i, X_k) = \text{Cov}(\hat{q}_i^T (X - \bar{X}), X_k) = \hat{q}_i^T S e_k = \hat{q}_i^T (\hat{\lambda}_i \hat{q}_i) = \hat{\lambda}_i \hat{q}_{ik}.$$

Step 2: Sample Correlation

Using $\text{Var}(\hat{Y}_i) = \hat{\lambda}_i$ and $\text{Var}(X_k) = S_{kk}$:

$$\hat{\rho}_{\hat{Y}_i, X_k} = \frac{\text{Cov}(\hat{Y}_i, X_k)}{\sqrt{\text{Var}(\hat{Y}_i) \text{Var}(X_k)}} = \frac{\hat{\lambda}_i \hat{q}_{ik}}{\sqrt{\hat{\lambda}_i S_{kk}}} = \frac{\hat{q}_{ik} \sqrt{\hat{\lambda}_i}}{\sqrt{S_{kk}}}.$$

Example: Population vs Sample PCA

Population Setup

Consider a bivariate random vector

$$X = (X_1, X_2)^T \sim N(\mu, \Sigma)$$

with

$$\mu = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \Sigma = \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix}.$$

$\text{Var}(X_1) = 1$, $\text{Var}(X_2) = 1$ and $\text{Cov}(X_1, X_2) = 0.8$

Goal

Compute population PCA theoretically, then simulate 5 observations to calculate sample PCA, and compare the two.

Step 1: Population PCA

Eigenvalues of Σ

Solve $\det(\Sigma - \lambda I) = 0$:

$$\begin{vmatrix} 1 - \lambda & 0.8 \\ 0.8 & 1 - \lambda \end{vmatrix} = (1 - \lambda)^2 - 0.64 = 0$$
$$\Rightarrow \lambda^2 - 2\lambda + 0.36 = 0 \Rightarrow \lambda_{1,2} = 1.8, 0.2$$

Eigenvectors of Σ

- For $\lambda_1 = 1.8$: $(\Sigma - 1.8I)q_1 = 0 \Rightarrow -0.8x + 0.8y = 0 \Rightarrow y = x$ Normalize: $q_1 = \frac{1}{\sqrt{2}}(1, 1)^T$
- For $\lambda_2 = 0.2$: orthogonal to $q_1 \Rightarrow q_2 = \frac{1}{\sqrt{2}}(-1, 1)^T$

Population Principal Components

$$Y_1 = \frac{1}{\sqrt{2}}(X_1 + X_2), \quad Y_2 = \frac{1}{\sqrt{2}}(-X_1 + X_2)$$

- Variances: $\text{Var}(Y_1) = 1.8$, $\text{Var}(Y_2) = 0.2$ - Covariance: 0 (uncorrelated)

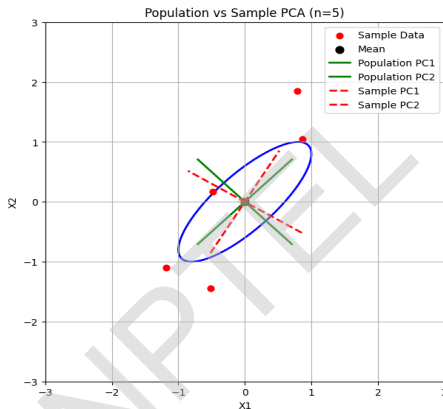
Step 2: Simulate Sample Observations

Simulation Procedure

- For sample sizes $n=5, 20, 100$: Generate $X_i \sim N(\mu, \Sigma)$, compute sample mean $\bar{X}$.
- Center the data $X_i - \bar{X}$, and compute sample covariance $S = \frac{1}{n-1} \sum (X_i - \bar{X})(X_i - \bar{X})^T$.

Observation: As n increases, the sample covariance S approaches Σ , so sample PCA converges to population PCA.

Step 3: Sample PCA for n=5



Sample Covariance and PCA (n=5)

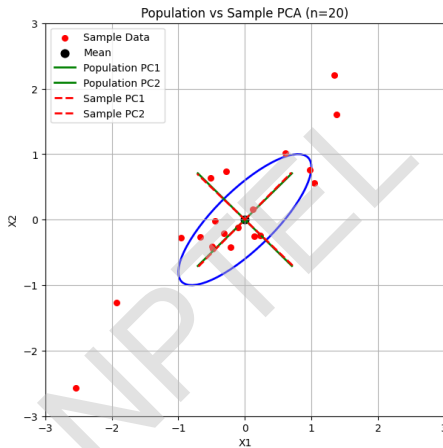
Sample covariance $S_5 \approx \begin{pmatrix} 0.7 & 0.56 \\ 0.56 & 1.3 \end{pmatrix}$.

Eigenvalues: $\hat{\lambda}_1 \approx 1.643, \hat{\lambda}_2 \approx 0.357$.

Eigenvectors: $\hat{q}_1 \approx (0.512, 0.859)^T, \hat{q}_2 \approx (-0.859, 0.512)^T$.

Interpretation: With a small sample, sample PCs differ noticeably from population PCs; variance and directions deviate due to high sampling variability.

Step 4: Sample PCA for n=20



Sample Covariance and PCA (n=20)

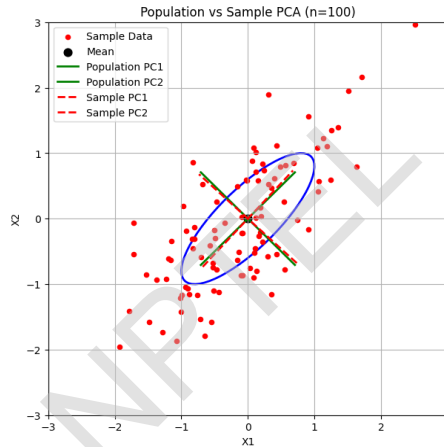
Sample covariance $S_{20} \approx \begin{pmatrix} 0.95 & 0.75 \\ 0.75 & 1.05 \end{pmatrix}$.

Eigenvalues: $\hat{\lambda}_1 \approx 1.68, \hat{\lambda}_2 \approx 0.32$.

Eigenvectors: $\hat{q}_1 \approx (0.71, 0.71)^T, \hat{q}_2 \approx (-0.71, 0.71)^T$.

Interpretation: Sample PCs are close to population PCs; small discrepancies due to sampling variability.

Step 5: Sample PCA for n=100



Sample Covariance and PCA (n=100)

Sample covariance $S_{100} \approx \begin{pmatrix} 0.99 & 0.80 \\ 0.80 & 1.01 \end{pmatrix}$.

Eigenvalues: $\hat{\lambda}_1 \approx 1.799, \hat{\lambda}_2 \approx 0.201$.

Eigenvectors: $\hat{q}_1 \approx (0.707, 0.707)^T, \hat{q}_2 \approx (-0.707, 0.707)^T$.

Observation: Sample PCA almost matches population PCA; variance and directions converge.

Step 6: Comparison Across Sample Sizes

Comparison of Population and Sample PCs

Sample size	$\hat{\lambda}_1$	$\hat{q}_1$
5	1.643	(0.512,0.859)
20	1.68	(0.71,0.71)
100	1.799	(0.707,0.707)
Population	1.8	(0.707,0.707)

Conclusion: As n increases, sample PCA converges to population PCA in both variance and direction, illustrating the law of large numbers for PCA.

References

- ① Richard Arnold Johnson, and W. Dean Wichern. Applied multivariate statistical analysis, 2002.
- ② I. T. Jolliffe, Principal Component Analysis, Springer New York, January 2002.
- ③ David C Lay, Judi McDonald, and Steven R Lay Linear Algebra and Its Applications, 6th edition, Pearson (2020).

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Module 10: Sample Principal Component Analysis

Lecture 03: Sample PCA for Covariance Matrix and Correlation Matrix



Sample Setup and Notation

Sample Observations

Let

$$X^{(1)}, X^{(2)}, \dots, X^{(n)} \in \mathbb{R}^p$$

be an i.i.d. sample from a given population.

Sample Mean and Covariance

Sample mean:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X^{(i)}.$$

Sample covariance matrix:

$$\hat{\Sigma} = \frac{1}{n-1} \sum_{i=1}^n (X^{(i)} - \bar{X})(X^{(i)} - \bar{X})^T.$$

Interpretation

Sample PCA analyzes variability present in a finite dataset as an empirical estimate of population PCA.

Goal of Sample PCA

Goal

Sample PCA seeks orthogonal linear combinations of the variables that capture the largest possible sample variance, using the sample covariance matrix $\hat{\Sigma}$.

Sample PCA: Setup

Centered Data

Define centered observations:

$$Z^{(i)} = X^{(i)} - \bar{X}.$$

Definition of Components

Sample principal components are:

$$\hat{Y}_k^{(i)} = q_k^T Z^{(i)}, \quad k = 1, \dots, p.$$

Goal

Find orthogonal directions q_k that:

- maximize the sample variance of $\hat{Y}_k$
- ensure the components are sample uncorrelated

Properties of Sample PCA

Variance

$$\text{Var}_n(\hat{Y}_k) = q_k^T \hat{\Sigma} q_k.$$

Uncorrelatedness

$$\text{Cov}_n(\hat{Y}_i, \hat{Y}_j) = q_i^T \hat{\Sigma} q_j = 0, \quad i \neq j.$$

Interpretation

The vectors q_k rotate the coordinate system so that the components $\hat{Y}_k$ are uncorrelated and ordered by decreasing sample variance.

General k -th Principal Component

Optimization Problem

$$q_k = \arg \max_{q \in \mathbb{R}^p} q^T \hat{\Sigma} q$$

subject to

$$q^T q = 1, \quad q^T q_j = 0 \text{ for all } j < k.$$

Component

$$\hat{Y}_k^{(i)} = q_k^T Z^{(i)}.$$

Interpretation

Each new component maximizes sample variance under the requirement of being uncorrelated with all previous components.

Sample Setup and Notation (Correlation PCA)

Standardized Sample Observations

Given i.i.d. observations

$$X^{(1)}, X^{(2)}, \dots, X^{(n)} \in \mathbb{R}^p,$$

define standardized variables for each coordinate j :

$$\tilde{X}_j^{(i)} = \frac{X_j^{(i)} - \bar{X}_j}{s_j},$$

where $\bar{X}_j$ is the sample mean and s_j is the sample standard deviation.

Centered and Standardized Vectors

Let

$$\tilde{Z}^{(i)} = \left(\tilde{X}_1^{(i)}, \dots, \tilde{X}_p^{(i)} \right)^T.$$

Sample Setup and Notation (Correlation PCA)

Sample Correlation Matrix

The sample correlation matrix is

$$\hat{R} = \frac{1}{n-1} \sum_{i=1}^n \tilde{Z}^{(i)} (\tilde{Z}^{(i)})^T.$$

Interpretation

PCA based on $\hat{R}$ analyzes relationships after standardizing all variables. This construction is fully analogous to PCA based on the sample covariance matrix $\hat{\Sigma}$, but removes scale effects.

Goal of Sample PCA (Correlation Matrix)

Goal

Sample PCA using $\hat{R}$ seeks orthogonal linear combinations of the standardized variables that capture the largest possible sample variance (as measured by $\hat{R}$).

Analogy to Covariance PCA

The only difference from covariance-based PCA is that the input matrix is

$$\hat{\Sigma} \rightarrow \hat{R}.$$

All optimization problems and properties remain the same.

Correlation PCA: Setup

Standardized Data

We use the standardized and centered observations

$$\tilde{Z}^{(i)} = X_{\text{standardized}}^{(i)}.$$

Definition of Components

The sample principal components (correlation-based) are

$$\hat{Y}_k^{(i)} = r_k^T \tilde{Z}^{(i)}, \quad k = 1, \dots, p.$$

Goal

Find orthogonal directions r_k that:

- maximize the sample variance of $\hat{Y}_k$ measured using $\hat{R}$,
- ensure the components are uncorrelated under $\hat{R}$.

Properties of Correlation-Based PCA

Variance of Component

$$\text{Var}_n(\hat{Y}_k) = r_k^T \hat{R} r_k.$$

This equals the k -th eigenvalue of $\hat{R}$.

Uncorrelatedness

For $i \neq j$:

$$\text{Cov}_n(\hat{Y}_i, \hat{Y}_j) = r_i^T \hat{R} r_j = 0.$$

Interpretation

The vectors r_k rotate the standardized coordinate system so that the components $\hat{Y}_k$ are uncorrelated and ordered by decreasing variance.

General k -th Principal Component (Correlation Matrix)

Optimization Problem

$$r_k = \arg \max_{r \in \mathbb{R}^p} r^T \hat{R} r$$

subject to

$$r^T r = 1, \quad r^T r_j = 0 \text{ for all } j < k.$$

Component

$$\hat{Y}_k^{(i)} = r_k^T \tilde{Z}^{(i)}.$$

Analogy to Covariance PCA

Identical optimization problem:

$$q_k = \arg \max q^T \hat{\Sigma} q \quad \text{and} \quad r_k = \arg \max r^T \hat{R} r.$$

The only change is replacing the covariance matrix by the correlation matrix.

Summary: Covariance PCA vs. Correlation PCA

Covariance PCA

Uses raw centered data. Sensitive to variable scales. Components satisfy:

$$\hat{Y}_k = q_k^T Z, \quad \text{Var}(\hat{Y}_k) = q_k^T \hat{\Sigma} q_k.$$

Correlation PCA (This Section)

Uses standardized data. Scale-invariant; each variable contributes equally. Components satisfy:

$$\hat{Y}_k = r_k^T \tilde{Z}, \quad \text{Var}(\hat{Y}_k) = r_k^T \hat{R} r_k.$$

Key Takeaway

Everything is structurally identical; replace $\hat{\Sigma}$ with $\hat{R}$ and use standardized data.

Example 1: Variables, Correlation Matrix, and Basic Statistics

Variables

We observe $p = 8$ blood–chemistry variables on $n = 72$ patients:

$$X = (\text{RBLOOD}, \text{PLATE}, \text{WBLOOD}, \text{NEUT}, \text{LYMPH}, \text{BILIR}, \text{SODIUM}, \text{POTASS})^T$$

Correlation Matrix R

	RBLOOD	PLATE	WBLOOD	NEUT	LYMPH	BILIR	SODIUM	POTASS
RBLOOD	1.000	0.290	0.202	-0.055	-0.105	-0.252	-0.229	0.058
PLATE	0.290	1.000	0.415	0.285	-0.376	-0.349	-0.164	-0.129
WBLOOD	0.202	0.415	1.000	0.419	-0.521	-0.441	-0.145	-0.076
NEUT	-0.055	0.285	0.419	1.000	-0.877	-0.076	0.023	-0.131
LYMPH	-0.105	-0.376	-0.521	-0.877	1.000	0.206	0.034	0.151
BILIR	-0.252	-0.349	-0.441	-0.076	0.206	1.000	0.192	0.077
SODIUM	-0.229	-0.164	-0.145	0.023	0.034	0.192	1.000	0.423
POTASS	0.058	-0.129	-0.076	-0.131	0.151	0.077	0.423	1.000

Key Observation

- Strongest correlation: $R_{\text{NEUT}, \text{LYMPH}} = -0.877$
- Indicates strong opposing behavior between neutrophils and lymphocytes.

Why Use Correlation PCA Instead of Covariance PCA?

Key Points

- Covariance PCA is based on the covariance matrix, which is sensitive to the scale of variables.
- When variables have different units or vastly different magnitudes, those with larger variance dominate the first principal components.
- Example: A variable measured in thousands can overshadow variables measured in small decimals, even if the smaller variables are highly correlated.
- Correlation PCA standardizes variables to unit variance before PCA, removing the effect of differing scales.
- This ensures that all variables contribute fairly, highlighting meaningful relationships rather than scale artifacts.

Covariance PCA vs Correlation PCA

Covariance PCA (S)

- Uses raw covariance matrix.
- PLATE has variance $41.253^2 \approx 1701$, dominating all other variables.
- First PC solves $\max_{r^\top r=1} r^\top S r$, explaining $\approx 98.6\%$ of total variance.
- Problem: PC reflects scale, not meaningful relationships.

Correlation PCA (R)

- Uses correlation matrix (standardized variables, variance = 1).
- Removes scale effects; all variables treated equally.
- First two PCs explain 34.9% and 19.1% of total variance.
- Eigenvectors (direction vectors of PCs) capture **patterns of association**:

$$q_1 = (0.2, 0.4, 0.4, -0.4, -0.4, -0.2, -0.2)$$

$$q_2 = (-0.4, -0.2, 0, 0.4, -0.4, 0.4, 0.6, 0.2)$$

Step 1: Scale Effect in Covariance PCA

Observation

- Variances differ drastically:

$$\text{var}(\text{PLATE}) \approx 1701, \quad \text{var}(\text{NEUT}) \approx 0.006, \quad \text{var}(\text{LYMPH}) \approx 0.005$$

- PLATE completely dominates the covariance matrix S .

Implication for PCA

- First PC aligns almost entirely with PLATE.
- Explains 98.6% of variance — misleading for patterns among other variables.

Step 2: Correlation PCA Highlights Relationships

Standardization

- All variables transformed to variance 1.
- Removes scale dominance (e.g., PLATE no longer dominates).

Key Patterns Captured

- Strong negative correlation: NEUT vs LYMPH (-0.877).
- Captures joint movement among other correlated variables (e.g., WBLOOD with NEUT).
- First two PCs now meaningful: explain 34.9% and 19.1% of variance.

Step 3: Direction Vectors of Correlation PCA

Eigenvectors of R

$$q_1 = (0.2, 0.4, 0.4, 0.4, -0.4, -0.4, -0.2, -0.2)$$

$$q_2 = (-0.4, -0.2, 0, 0.4, -0.4, 0.4, 0.6, 0.2)$$

Interpretation

- q_1 : WBLOOD & NEUT increase together, LYMPH & BILIR decrease — main opposing pattern.
- q_2 : SODIUM/POTASS oppose NEUT/LYMPH — secondary pattern.

Step 4: Converting PCA Directions to Original Units

Reason for Conversion

- Eigenvectors of R are for standardized variables.
- To interpret in original units, rescale using eigenvalues:

$$\tilde{q}_1 = D^{-1/2} q_1$$

where D is the diagonal matrix of eigenvalues of R .

Resulting First Direction in Original Units

Variable	$\tilde{q}_{1,i}$
RBLOOD	0.54
PLATE	0.0097
WBLOOD	0.21
NEUT	5.20
LYMPH	-5.63
BILIR	-0.10
SODIUM	-0.073
POTASS	-0.67

Step 5: Key Lessons from This Example

Interpretation

- First PC dominated by NEUT & LYMPH — captures main opposing pattern.
- PLATE has negligible contribution after standardization — confirms correlation PCA is scale-invariant.

Covariance PCA

- Dominated by variables with largest variance (PLATE).
- First PC explains most variance but is ****misleading**** biologically.

Correlation PCA

- Highlights true patterns among variables regardless of scale.
- First PC captures NEUT–LYMPH opposition (biologically meaningful).
- Second PC captures SODIUM/POTASS vs immune variables (secondary pattern).
- Converting eigenvectors using eigenvalues allows interpretation in original units.

Biological Insight

- PCA reveals structured variation among immune cells and electrolytes.
- Correlation PCA is recommended when variables have different scales.

References

- ① Richard Arnold Johnson, and W. Dean Wichern. Applied multivariate statistical analysis, 2002.
- ② I. T. Jolliffe, Principal Component Analysis, Springer New York, January 2002.
- ③ David C Lay, Judi McDonald, and Steven R Lay Linear Algebra and Its Applications, 6th edition, Pearson (2020).

Thank you





SWAYAM NPTEL COURSE ON STATISTICAL FOUNDATION FOR BIG DATA ANALYSIS

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Module 10: Sample Principal Component Analysis

Lecture 04: Examples of Sample PCA



Sample PCA: Core Ideas

Setup

- Observations: $X^{(1)}, \dots, X^{(n)} \in \mathbb{R}^p$.
- Use sample mean and covariance to summarize variability.

Goal of PCA

- Find orthogonal directions q_k that maximize sample variance.
- Principal components: $\hat{Y}_k = q_k^T (X - \bar{X})$.
- Components are uncorrelated and ordered by decreasing variance.

Interpretation

PCA rotates coordinates to reveal dominant patterns of variation.

Covariance PCA vs Correlation PCA

Covariance PCA

- Based on covariance matrix $\hat{\Sigma}$.
- Highly sensitive to variable scale.
- Large-variance variables dominate the first PCs.

Correlation PCA

- Uses standardized data $\rightarrow$ correlation matrix $\hat{R}$.
- Each variable has unit variance.
- Removes scale effects; all variables contribute equally.

Key Point

Optimization is identical; only substitute

$$\hat{\Sigma} \rightarrow \hat{R}.$$

Example 1 (From the Previous Lecture): Why Correlation PCA?

Data

8 blood-chemistry variables measured on 72 patients.

Problem with Covariance PCA

- PLATE has extremely large variance.
- First PC becomes dominated by PLATE.
- Explains $\approx 98\%$ of total variance but is not meaningful.

Using Correlation PCA

- Standardization puts variables on equal footing.
- Reveals true relationships:

Strong opposition between NEUT and LYMPH (-0.877).

- First two PCs capture actual biological structure.

Example 1 (From the Previous Lecture): Meaningful Components from Correlation PCA

First Principal Component

- Captures dominant NEUT–LYMPH opposition.
- WBLOOD and BILIR also move systematically with these variables.

Second Principal Component

- SODIUM and POTASS contrast with immune-related variables.
- Reveals secondary physiological pattern.

Key Lessons

- Covariance PCA can be misleading when variable scales differ.
- Correlation PCA highlights genuine relationships among variables.
- Most meaningful interpretation in Example 1 comes from correlation PCA.

Example 2: Reflex Measurements at 10 Body Sites

Sample Data

Measurements of reflex strength at ten body sites (variables $V1, \dots, V10$) were collected for $n = 143$ individuals. Let the observations be

$$X^{(1)}, \dots, X^{(n)} \in \mathbb{R}^{10}, \quad \bar{X} = \frac{1}{n} \sum_{i=1}^n X^{(i)}.$$

Sample Correlation Matrix R

	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10
V1	1.00	0.98	0.60	0.71	0.55	0.55	0.38	0.25	0.22	0.20
V2	0.98	1.00	0.62	0.73	0.57	0.57	0.40	0.28	0.21	0.19
V3	0.60	0.62	1.00	0.88	0.61	0.56	0.48	0.42	0.19	0.18
V4	0.71	0.73	0.88	1.00	0.68	0.68	0.53	0.47	0.23	0.21
V5	0.55	0.57	0.61	0.68	1.00	0.97	0.33	0.27	0.16	0.16
V6	0.55	0.57	0.56	0.68	0.97	1.00	0.33	0.27	0.19	0.39
V7	0.38	0.40	0.48	0.53	0.33	0.33	1.00	0.90	0.40	0.40
V8	0.25	0.28	0.42	0.47	0.27	0.27	0.90	1.00	0.41	0.40
V9	0.22	0.21	0.19	0.23	0.16	0.19	0.40	0.41	1.00	0.94
V10	0.20	0.19	0.18	0.21	0.16	0.39	0.40	0.40	0.94	1.00

Sample Principal Components

Eigen-Decomposition of R

Let

$$R\hat{q}_j = \hat{\lambda}_j \hat{q}_j, \quad \hat{\lambda}_1 \geq \dots \geq \hat{\lambda}_{10}.$$

The sample PCs are

$$\hat{Y}_j = \hat{q}_j^\top (X - \bar{X}), \quad j = 1, \dots, 10.$$

Eigenvectors $\hat{q}_j$ (Principal Component Coefficients)

	$\hat{q}_1$	$\hat{q}_2$	$\hat{q}_3$	$\hat{q}_4$	$\hat{q}_5$	$\hat{q}_6$	$\hat{q}_7$	$\hat{q}_8$	$\hat{q}_9$	$\hat{q}_{10}$
V1	0.3	-0.2	0.2	-0.5	0.3	0.1	-0.1	-0.0	-0.6	0.2
V2	0.4	-0.2	0.2	-0.5	0.3	0.0	-0.1	-0.0	0.7	-0.3
V3	0.4	-0.1	-0.1	-0.0	-0.7	0.5	-0.2	0.0	0.1	0.1
V4	0.4	-0.1	-0.1	-0.4	-0.7	0.3	-0.0	-0.1	-0.1	-0.1
V5	0.3	-0.2	0.1	0.5	0.2	0.2	-0.0	-0.1	-0.2	-0.6
V6	0.3	-0.2	0.2	0.5	0.2	-0.1	-0.0	0.1	0.2	0.6
V7	0.3	0.3	-0.5	-0.0	0.2	0.3	0.7	0.0	-0.0	0.0
V8	0.3	0.3	-0.5	0.1	0.2	-0.2	-0.7	-0.0	-0.0	-0.0
V9	0.2	0.5	0.4	0.0	-0.1	0.0	-0.0	0.7	-0.0	-0.1
V10	0.2	0.5	0.4	0.0	-0.1	0.0	-0.0	-0.7	0.0	0.0

Interpretation of PCA Structure

Percentage of Total Variation Explained

PC	1	2	3	4	5	6	7	8	9	10
%Var	52.3	20.4	11.0	8.5	5.0	1.0	0.9	0.6	0.2	0.2

Key Structural Interpretations

- The correlation matrix shows **five highly correlated variable pairs**: $\{V1, V2\}, \{V3, V4\}, \{V5, V6\}, \{V7, V8\}, \{V9, V10\}$.
- Because within-pair correlations are very high, **PCs 6–10** are mainly within-pair contrasts with very small variances.
- **PC1** has all positive weights → “overall reflex strength” component.
- **PC2** contrasts the two groups: $\{V1, \dots, V6\}$ vs. $\{V7, \dots, V10\}$.
- **PC3–PC5** refine group-based contrasts: PC3 contrasts pairs in the smaller group; PC4–PC5 contrast pairs within the larger group.
- Remarkably, all 10 PCs are interpretable due to the unusually clean correlation structure.

Why This Example Matters: Blocked Correlation Structure

Key Issue

- The ten reflex–measurement variables form *five very strongly correlated pairs*.
- Each pair corresponds to reflexes measured at two anatomically related body sites.
- The correlation matrix therefore has a clear *block–pair structure*.

Consequences for PCA

- PCA will first extract variation that is common across all variables.
- Then it will identify contrasts between the strongly correlated groups.
- Finally, it will isolate small within-pair differences.

Step 1: Understanding the Correlation Matrix

Observed Structure

- The matrix shows clusters with extremely high correlations:

$$\text{corr}(V1, V2) = 0.98, \quad \text{corr}(V3, V4) = 0.88, \quad \text{corr}(V5, V6) = 0.97,$$

$$\text{corr}(V7, V8) = 0.90, \quad \text{corr}(V9, V10) = 0.94.$$

- These define *natural pairs*: reflex measurements taken at two sides of the body or two parallel anatomical positions.

Implications for PCA

- The high correlations mean each pair behaves almost like a single variable.
- The effective dimension is therefore much smaller than ten.
- PCA will treat each pair as a nearly collinear 2D cluster.

Step 2: Eigenstructure of the Correlation Matrix

General Behaviour of Eigenvectors

- When variables form strong pairs, the first components capture variation common across all pairs.
- Later components capture contrasts *within* pairs because each pair produces a small eigenvalue direction.
- The eigenvectors reflect the block structure very clearly.

Observed Eigenvectors $\hat{q}_1, \dots, \hat{q}_{10}$

- $\hat{q}_1$ has all positive entries: *global overall reflex strength*.
- $\hat{q}_2$ assigns negative weights to $\{V1, \dots, V6\}$ and positive weights to $\{V7, \dots, V10\}$, so it contrasts the two major clusters.
- $\hat{q}_3, \hat{q}_4, \hat{q}_5$ refine contrasts among clusters.
- $\hat{q}_6$ to $\hat{q}_{10}$ contain opposite signs within each pair, indicating *within-pair differences*.

Step 3: Interpretation of Variance Explained

Explained Variation

PC	1	2	3	4	5	6	7	8	9	10
%Var	52.3	20.4	11.0	8.5	5.0	1.0	0.9	0.6	0.2	0.2

Meaning

- Over half the variation (PC1) corresponds to an *overall signal* common to all ten reflex sites.
- The next 20.4% (PC2) represents a *large contrast* between the two major anatomical groups.
- PCs 3–5 capture smaller inter-group contrasts.
- PCs 6–10 have very small variance because they correspond to *differences within each strongly correlated pair*.

Step 4: Detailed Interpretation of Principal Components

PC1: Overall Reflex Strength

- All coefficients have the same sign.
- Indicates a general “strength of reflexes” component.
- Individuals with high values of PC1 have high reflex strength across all 10 sites.

PC2: Major Group Contrast

- Opposing signs between the first six variables and the last four.
- Describes systematic difference between two anatomical regions.
- Large PC2 scores indicate reflexes stronger in one anatomical group than the other.

Step 5: Higher-Order Components

PC3–PC5: Subgroup Contrasts

- These components refine contrasts among the groups.
- PC3 emphasises variation within the smaller groups.
- PC4–PC5 emphasise variation inside the larger clusters.
- They capture mid-level anatomical differences.

PC6–PC10: Within-Pair Oppositions

- Each strongly correlated pair contributes one near-zero eigenvalue.
- These PCs isolate small left–right or dual-site differences.
- Signs in each pair are opposite: e.g. (V1,V2), (V3,V4), (V5,V6), (V7,V8), (V9,V10).
- These directions explain extremely small variation (less than 1.0% each).

Final Conclusions from the Example

From the Correlation Matrix

- The reflex measurements form well-defined, highly correlated anatomical pairs.
- This produces a natural dimensionality far below 10.

From the PCA Results

- PC1 captures overall reflex strength across the entire body.
- PC2 separates major anatomical regions.
- PCs 3–5 capture medium-scale contrasts between subgroups.
- PCs 6–10 capture only minor left–right or dual-site asymmetries.

Practical Message

- The PCA structure is exceptionally interpretable because of the clean block correlations.
- High-correlation pairs create near-collinear directions, leading directly to low-variance components.

References

- ① Richard Arnold Johnson, and W. Dean Wichern. Applied multivariate statistical analysis, 2002.
- ② I. T. Jolliffe, Principal Component Analysis, Springer New York, January 2002.
- ③ David C Lay, Judi McDonald, and Steven R Lay Linear Algebra and Its Applications, 6th edition, Pearson (2020).

Thank you





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Module 10: Sample Principal Component Analysis

Lecture 05: Inference on Principal Component Analysis



Motivation for MLE in PCA

Population Setting

Let

$$X_1, X_2, \dots, X_n \sim N_p(\mu, \Sigma)$$

be i.i.d. observations from a population with unknown mean μ and covariance Σ .

Goal of PCA

- Identify directions $q_1, \dots, q_p$ in which variance is maximized
- Estimate corresponding population variances (eigenvalues) $\lambda_1 \geq \dots \geq \lambda_p$
- Make inference about the true population structure, not just the sample

Challenge

- Sample covariance $\hat{\Sigma}$ is a finite-sample estimate of Σ
- Without a probabilistic model, there is no principled way to quantify uncertainty or test hypotheses (we also assume normality).

Maximum Likelihood Estimation in PCA

Likelihood Function

For multivariate normal data:

$$\ell(\mu, \Sigma) = -\frac{n}{2} \log |\Sigma| - \frac{1}{2} \sum_{i=1}^n (X_i - \mu)^T \Sigma^{-1} (X_i - \mu)$$

MLEs

- Mean: $\hat{\mu} = \bar{X}$
- Covariance: $\hat{\Sigma}_{\text{MLE}} = \frac{1}{n} \sum_{i=1}^n (X^{(i)} - \bar{X})(X^{(i)} - \bar{X})^T$
- Sample principal directions q_k and eigenvalues $\hat{\lambda}_k$ are MLEs of the population quantities
- MLE provides statistically optimal estimates under the Gaussian assumption
- Allows derivation of asymptotic distributions for eigenvalues and eigenvectors

MLE of Eigenvalues in PCA

Eigen-Decomposition of the MLE Covariance

Let

$$\hat{\Sigma}_{\text{MLE}} = Q \hat{\Lambda} Q^{\top}, \quad \hat{\Lambda} = \text{diag}(\hat{\lambda}_1, \dots, \hat{\lambda}_p).$$

MLE of the Population Eigenvalues

$$\hat{\lambda}_k = q_k^{\top} \hat{\Sigma}_{\text{MLE}} q_k = \frac{1}{n} \sum_{i=1}^n (q_k^{\top} (X_i - \bar{X}))^2$$

- Each $\hat{\lambda}_k$ is the MLE of the population eigenvalue λ_k
- It equals the sample variance in the k -th principal direction
- Comes directly from maximizing the Gaussian likelihood

Why Hypothesis Testing for Dimensionality is Required in PCA

Motivation

- After obtaining sample principal components, we want to know which components are meaningful
- Without testing, one may include components that capture only sampling noise

Population Hypothesis for Dimensionality

- Test if the last $p - k$ population components contribute equally:

$$H_0 : \lambda_{k+1} = \cdots = \lambda_p$$

- Under H_0 , the bottom components have no preferred direction; they do not carry additional information
- Rejecting H_0 indicates additional meaningful population structure beyond the first k components

Connection Between MLE and Hypothesis Testing for Dimensionality in PCA

Step 1: Use MLE for Estimation

- Compute MLE covariance $\hat{\Sigma}_{\text{MLE}}$
- Compute its eigenvalues $\hat{\lambda}_1 \geq \dots \geq \hat{\lambda}_p$ and eigenvectors $q_1, \dots, q_p$
- These are statistically principled estimates of the population quantities

Step 2: Test for Dimensionality

- Focus on bottom block $\hat{\lambda}_{k+1}, \dots, \hat{\lambda}_p$
- Form a test statistic (e.g., log-ratio of arithmetic and geometric means) to measure departure from equality
- Reference distribution under H_0 is chi-squared with degrees of freedom $\frac{1}{2}(p - k)(p - k - 1)$

Why Both Are Needed

- MLE ensures estimates reflect population parameters
- Hypothesis testing provides a formal decision rule to identify the number of meaningful components
- Together, they allow PCA to move from descriptive analysis to statistical inference

Maximum Likelihood PCA: Model

Population Model

Assume each observation is i.i.d. from a multivariate normal:

$$X_1, X_2, \dots, X_n \sim N_p(\mu, \Sigma).$$

Log-Likelihood Function

The log-likelihood (up to constant) is:

$$\ell(\mu, \Sigma) = -\frac{n}{2} \log |\Sigma| - \frac{1}{2} \sum_{i=1}^n (X_i - \mu)^T \Sigma^{-1} (X_i - \mu).$$

MLEs

Maximizing w.r.t. μ and Σ gives:

$$\hat{\mu} = \bar{X}, \quad \hat{\Sigma}_{\text{MLE}} = \frac{1}{n} \sum_{i=1}^n (X^{(i)} - \bar{X})(X^{(i)} - \bar{X})^T.$$

MLE of Principal Components

Eigen-Decomposition of $\hat{\Sigma}_{\text{MLE}}$

$$\hat{\Sigma}_{\text{MLE}} = Q\Lambda Q^T, \quad Q = [q_1, \dots, q_p], \quad \Lambda = \text{diag}(\hat{\lambda}_1, \dots, \hat{\lambda}_p)$$

with $\hat{\lambda}_1 \geq \dots \geq \hat{\lambda}_p \geq 0$.

MLE Interpretation

$\hat{\lambda}_k$ = MLE of population variance along q_k

q_k = MLE of population principal direction

Sample PCA is exactly the MLE of the Gaussian covariance structure

Sample Component Scores

$$\hat{Y}_k^{(i)} = q_k^T Z^{(i)}, \quad Z^{(i)} = X^{(i)} - \bar{X}.$$

Testing Population PCA Dimensionality

Population Hypothesis

Test if the last $p - k$ population components contribute equally:

$$H_0 : \lambda_{k+1} = \dots = \lambda_p \text{ for population covariance } \Sigma.$$

Step 1: Sample Eigenvalues

Compute eigenvalues of $\hat{\Sigma}_{MLE}$:

$$\hat{\lambda}_1 \geq \dots \geq \hat{\lambda}_p$$

and focus on the bottom block $\hat{\lambda}_{k+1}, \dots, \hat{\lambda}_p$.

Step 2: Test Statistic

Arithmetic and Geometric Means

$$m = p - k, \quad \bar{\lambda} = \frac{1}{m} \sum_{j=k+1}^p \hat{\lambda}_j, \quad g = \left(\prod_{j=k+1}^p \hat{\lambda}_j \right)^{1/m}.$$

Log-Ratio Measure

$$L = \sum_{j=k+1}^p \ln \left(\frac{\hat{\lambda}_j}{\bar{\lambda}} \right)$$

- $L = 0$ if all bottom eigenvalues are equal - $L > 0$ indicates unequal eigenvalues

Finite-Sample Corrected Statistic

$$T = - \left(n - 1 - \frac{2p + 5}{6} \right) L = - \left(n - 1 - \frac{2p + 5}{6} \right) \sum_{j=k+1}^p \ln \left(\frac{\hat{\lambda}_j}{\bar{\lambda}} \right)$$

Step 3: Reference and Decision (Critical Region)

Reference Distribution

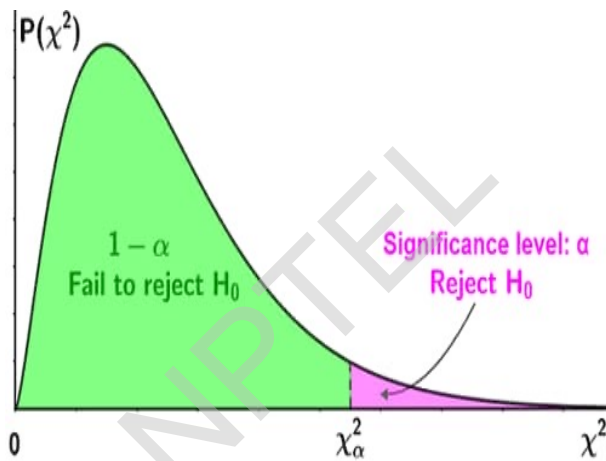
- Under H_0 (population eigenvalues equal):

$$T \sim \chi^2_{\text{df}}, \quad \text{df} = \frac{1}{2}m(m-1) = \frac{1}{2}(p-k)(p-k-1)$$

- Distribution is asymptotic.

Decision Rule (Critical Region)

- Define critical value $\chi^2_{\text{df}, 1-\alpha}$ for significance level α
- Reject H_0 if $T > \chi^2_{\text{df}, 1-\alpha}$
- Otherwise, do not reject H_0



Interpretation

- If $T > \chi^2_{df, 1-\alpha}$ then additional meaningful components beyond k
- If $T \leq \chi^2_{df, 1-\alpha}$ then last $p - k$ directions are indistinguishable from noise

References

- ① Richard Arnold Johnson, and W. Dean Wichern. Applied multivariate statistical analysis, 2002.
- ② I. T. Jolliffe, Principal Component Analysis, Springer New York, January 2002.
- ③ David C Lay, Judi McDonald, and Steven R Lay Linear Algebra and Its Applications, 6th edition, Pearson (2020).

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